



In the Future Combat System, each person is a network with routing capability to everyone else"

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dependant on ambient geography. Moreover, a commander receiving information from various sources has to collate that information and make a mental picture of the situation. There may sometimes be conflicting information. Many a time, it so happens that a commander doesn't have the precise location of friendly or enemy troops. Ambiguous states such as these can hamper decision-making. In such situations, a commander has to rely on experience and gut-feeling. It's sometimes a bit like gambling. With lives.

Taking an example of air combat, radars are a must for any fighter aircraft. But there are drawbacks associated with the use of radar. First, fighter aircrafts have a very small radar range compared to ground-based radars. Second, switching on your radar gives away your position to the enemy. So during most air combat scenarios, fighters keep their radars off, being guided by ground-based radars or AWACS (Airborne Warning and Control Systems): these are aircraft with huge radars of extremely long ranges mounted on them. In addition to pilots, AWACS have controllers who have the details about all aircraft within their radar cover. These controllers can inform friendly pilots about enemy movement and help them plan their tactics.

Now, in this conventional scenario, a radar operator or an AWACS controller uses radio to communicate with the pilots. The pilots have to create a mental picture of their position relative to the enemy aircraft and plan their approach. Air combat is highly dynamic due to the high speeds involved, and it takes time for a pilot to create a mental picture. If, during this time, the enemy aircraft make a sudden change in their path, the radar operator will need to update the pilots about it. The pilot will need to scrap his earlier idea and start from scratch.

In NCW, secure data links on the aircraft and ground station would be installed, and they would be linked via satellite. In this case, the radar operator can share visuals in real-time. Instead of creating a mental picture, pilots will have an exact idea of enemy locations, and can plan their approach quicker to gain the upper hand.

Iridium

Remember Iridium? The satellite phone service? It boasted of global coverage provided by a "constellation" of 77 satellites, with only 66 having been commissioned. During the recent Iraq conflict, US and UK forces made good use

of Iridium not only to increase combat efficiency but also to boost morale. The US forces transmitted voice and data for tactical purposes using the service, while the British military let soldiers talk to their families. Both militaries also used commercial GSM networks during the conflict—and are still using it. They use the Sectéra Secure Wireless Phone, which includes a clip-in security module that uses cryptographic algorithms to ensure secure communication.

These are just two instances of commercial setups being used in warfare—and we're likely to see more of this. Like the use of RFID tags to track men and materials. Like the use of smart cards, using which a soldier would have access to his assigned weapons, and by which an injured soldier's medical records would be instantly accessible.

Talking To Each Other

Operation Red Dawn to capture Saddam Hussein saw an integration of lots of new technologies. Thanks to the integrated system, intelligence reports were analysed and forces

prepared within a few hours. Looking at how tense and time-bound this decision was, an extremely short time from tip-off to capture was no small matter.

Each component involved in this integrated system was not exemplary on its own, and had existed for quite some time. US forces have been using UAVs and satellites to gather information for quite some time now. But this time, all the data was communicated in real time. It was sent to the US, front line units, and to the commanders who made the decisions.

Says John Hillen, senior Vice President and Director of the defense and intelligence group at American Management Systems (AMS), an IT

consultancy in Virginia, "There were a lot of new technologies out there, but none of them was particularly dramatic. It was the stitching together of all of these things in real-time and their integration into the operation."

As recently as in the first Gulf War, air commanders had to print hundreds of pages of "air tasking orders" to be sent out to various air units. These printouts would then be sent to airbases or aircraft carriers, where they were further sorted out to be given to pilots to inform them about their tasks. In the second Gulf war, IT had already "invaded" the battlefield. A single Humvee became a mobile data centre. Intelligence updates provided by various sensors were fed into a Unix-based work-

